<Academia de Código>

26/01/23

SUMMARIZER

<Rodrigo Caçorino>
Code Cadet

#81_Bootcamp {lisboa}

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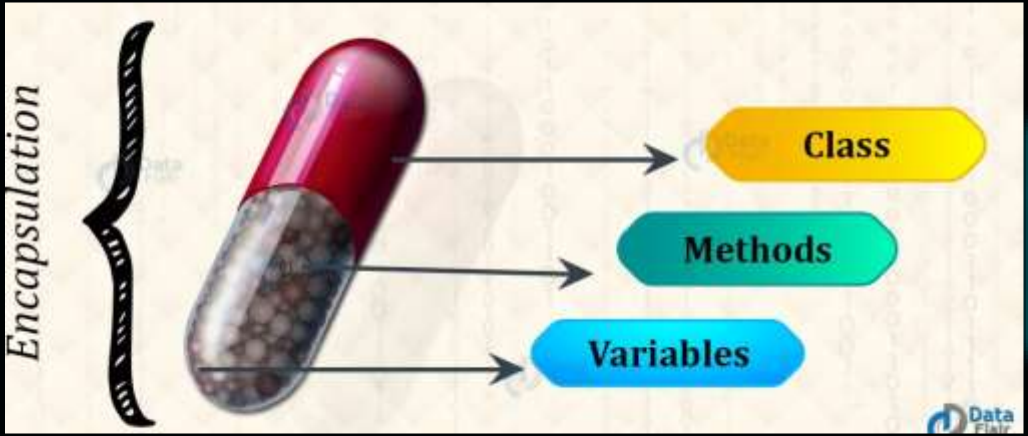
Encapsulation



01

DATA HIDING

CONTROL OVER THE DATA

EASY TO TEST

```java
/**
* Calculator.java
*/

public class Calculator {

    // hiding the data from the "outside"
    private String brand;
    private String color;

    public Calculator(String brand, String color) {
        this.brand = brand;
        this.color = color;
    }

    // GETTER
    public String getBrand() {
        return this.brand;
    }

    // SETTER
    public void setBrand(String brand) {
        this.brand = brand;
    }
}
```

```java
class Person {
  private int age;

  public void setAge(int age) {
    if (age >= 0) {
      this.age = age;
    }
  }
}
```

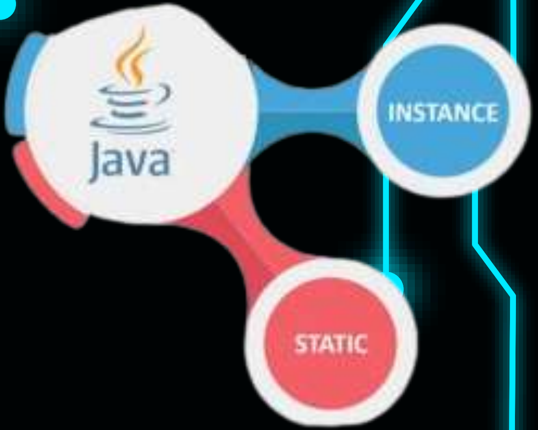
01

ACCESSING THE DATA

```java
/**
* Main.java
*/

public static void main(String[] args) {

    // using the new constructor
    Calculator calculator = new Calculator("Casio", "Grey");

    // using the SETTER
    calculator.setBrand("Texas Instruments");

    // using the GETTER
    String brand = calculator.getBrand();

    int result = calculator.add(3, 4);

    System.out.println(result);
}
```

02

Static vs Instance

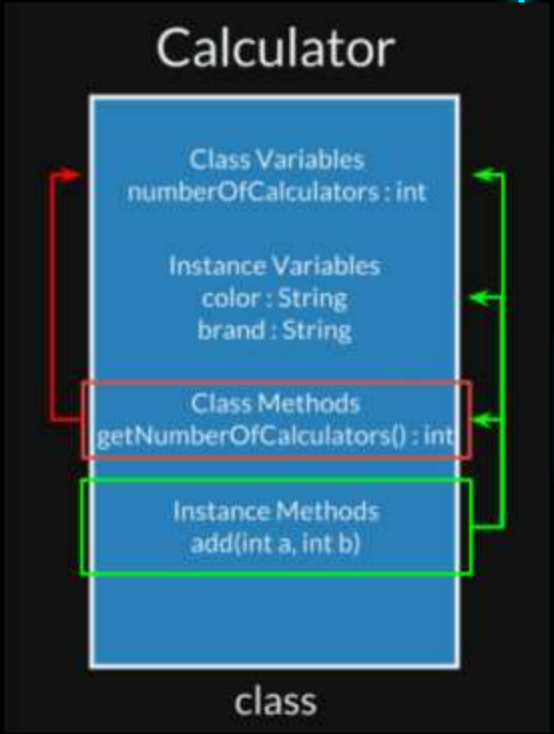


02

STATIC VARIABLES	INSTANCE VARIABLES
We can access them using class names.	Can accessed only using objects.
We can access them using static and instance methods.	Can be accessed only using instance methods
Static variables have global scope.	They have local scope

02 Static vs Class Methods

- We can access static methods from outside of the class in which they are defined;
- These methods can only access static variables of other classes as well as their own class;
- Can be used to create utility classes that contain general purpose methods.



03

Pair Programming

WHY?

03

HOW?

- PLAN TOGETHER
- COMMUNICATE
- AVOID BACKSEAT DRIVING
- SWITCH FREQUENTLY


WHAT IS BACKSEAT GAMING
STOP

04

Exercise - Guess a Number

(Using Pair Programming)

04

Main.java | NumberGenerator.java | NumberMaster.java | Player.java

```java
package org.academiadecodigo.bootcamp;

2 usages
public class NumberGenerator {

    2 usages
    public static int generateNumber(){

        return (((int) (Math.random() * (10))) + 1);

    }
}

```

Example

```java
public class Main {
  public static void main(String[] args) {
    double myDouble = 9.78d;
    int myInt = (int) myDouble; // Manual casting: double to int
```

04

Main.java | NumberGenerator.java | NumberMaster.java | Player.java

```java
package org.academiadecodigo.bootcamp;

2 usages
public class Player {
    2 usages
    private int currentGuessedNumber;

    2 usages
    public int getGuessedNumber() {

        return currentGuessedNumber;
    }
    1 usage
    public void getRandomNumber(){

        currentGuessedNumber = NumberGenerator.generateNumber();
    }
}
```

04

Main.java | NumberGenerator.java | NumberMaster.java | Player.java

```java
package org.academiadecodigo.bootcamp;

// 2 usages
public class NumberMaster {
    // 2 usages
    private int correctNumber = NumberGenerator.generateNumber();

    // 3 usages
    Player player1 = new Player();

    // 1 usage
    public void startGame(){

        while (correctNumber != player1.getGuessedNumber()) {

            System.out.println("Wrong. The correct number wasn't " + player1.getGuessedNumber() + ". Keep trying.");
            player1.getRandomNumber();
        }
        System.out.println("You guessed the correct number. It was: " + correctNumber + " congratz.");
```

04

```java
package org.academiadecodigo.bootcamp;

2 usages
public class NumberMaster {
    1 usage
    private int correctNumber = NumberGenerator.generateNumber();

    2 usages
    Player player1 = new Player();

    1 usage
    public void startGame(){

        if (getCorrectNumber() == player1.getGuessedNumber()) {
            System.out.println("You guessed the correct number. It was: " + correctNumber + " congratz.");
        }
        else {
            System.out.println("Wrong. The correct number wasn't " + player1.getGuessedNumber() + ". Keep trying.");
        }

        /* while (correctNumber != player1.getGuessedNumber()) {

            System.out.println("Wrong. The correct number wasn't " + player1.getGuessedNumber() + ". Keep trying.");
            player1.getRandomNumber();
        }
        System.out.println("You guessed the correct number. It was: " + correctNumber + " congratz."); */

```

04

Main.java | NumberGenerator.java | NumberMaster.java | Player.java

```java
package org.academiadecodigo.bootcamp;

no usages
public class Main {

    no usages
    public static void main(String[] args) {


        NumberMaster master = new NumberMaster();

        master.startGame();
    }
}
```

```
Wrong. The correct number wasn't 2. Keep trying.
Wrong. The correct number wasn't 3. Keep trying.
Wrong. The correct number wasn't 10. Keep trying.
Wrong. The correct number wasn't 10. Keep trying.
Wrong. The correct number wasn't 9. Keep trying.
Wrong. The correct number wasn't 2. Keep trying.
Wrong. The correct number wasn't 4. Keep trying.
Wrong. The correct number wasn't 7. Keep trying.
Wrong. The correct number wasn't 9. Keep trying.
Wrong. The correct number wasn't 2. Keep trying.
You guessed the correct number. It was: 1. Congratz.

Process finished with exit code 0
```

THANKS!